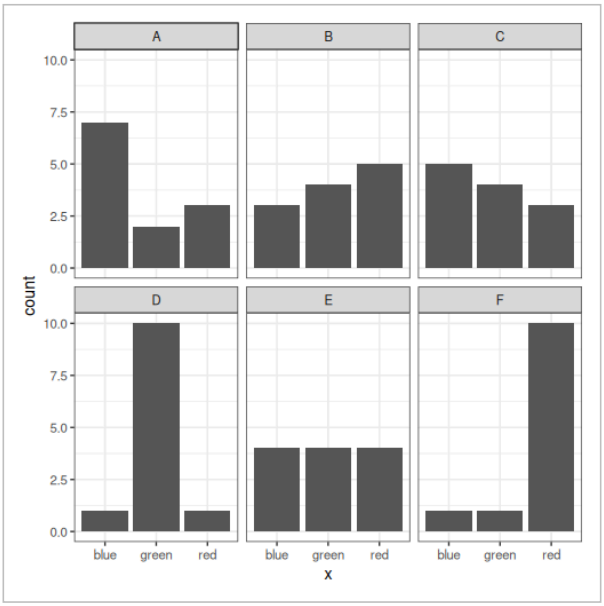
Warmup/Attendance: [Week 10: Wednesday 10/28/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

1. For any 2 plots, which pair would have the *greatest* chi-squared distance between them? (consider one of them the "observed" and the other the "expected"). **Once you finish answering this question, wait for your instructor before beginning the the remainder of the questions**!



2. Getting In Shape

Wait for your instructor to display two shapes. Which of the two names belongs to the shape on the **left**? Answer below. You have *15 seconds!*

Now, wait for your instructor to collect the responses from the entire class to this question. Then, you may begin questions 3 and 4 on the back side of this worksheet.

3. The Null Hypothesis

- Let p_k be the probability that a person selects Kiki for the shape on the left.
- Let $\hat{p}_k$ be the sample proportion of people that selected Kiki for the shape on the left.

Write down an equation for the null hypothesis that corresponds to the notion the link between names and shapes is arbitrary.

4. Approximate the null distribution

Our technique: simulate data from a world in which the null is true, then calculate the test statistic on the simulated data.

Which simulation method(s) align with the null hypothesis and our data collection process of simulating from a world where the null is true?

- Make a box with two cards, one that says "Kiki", the other that says "Bouba". Draw out n cards (the number of people who filled out the poll) with replacement, then compute $\hat{p}_k$.
- Make a box with two cards, one that says "Kiki", the other that says "Bouba". Draw n cards without replacement, then compute $\hat{p}_k$.
- Make a box with n cards, half of which say "Kiki", the other half of which say "Bouba". Draw n cards with replacement and calculate $\hat{p}_k$.
- Make a box with n cards, half of which say "Kiki", the other half of which say "Bouba". Draw n cards without replacement and calculate $\hat{p}_k$.
- Take a fair coin and assign Heads to "Kiki" and Tails to "Bouba". Flip the coin n times and calculate $\hat{p}_k$.